Shibin Jaison

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- ✓ I am an **AI & ML Engineer** with an M.E. in Computer Science and a B.E. in CSE (AI & ML), specializing in deep learning, computer vision, and robotics, with 1.5+ years of internship experience.
- ✓ I am skilled in applied artificial intelligence, computer vision, and robotics, enabling me to develop innovative solutions at the corporate level.
- ✓ With a strong technical foundation, I excel at building machine learning models end-to-end, handling everything from data preprocessing and training to hyperparameter tuning and testing.
- ✓ I have demonstrated expertise in designing and implementing deep learning models, including Graph Convolutional Neural Networks (GCNNs) for 3D reconstruction and hand detection systems for Human–Computer Interaction (HCI).

Skills

Programming Languages
Python • SQL • HTML/CSS

ML/DL Frameworks & Libraries
OpenCV • MediaPipe • TensorFlow • TensorFlow Lite • NumPy • Pandas • Keras • Scikit-Learn

Computer Vision & Specializations
Object Detection • Hand Detection & Tracking • Image Processing • Real-time Gesture Recognition • 3D Reconstruction • Face Detection • Semantic Segmentation • Feature Extraction

Deep Learning Architectures
Convolutional Neural Networks (CNN) • Graph Convolutional Neural Networks (GCNN) • Hand Detection Models • Recurrent Neural Networks (RNN) • Transformer basics

Tools, Platforms & Frameworks
Google Colab • Jupyter Notebook • Arduino IDE • GitHub • OpenCV

Technical Domains
Machine Learning • Computer Vision • Real-time Processing • 3D Mesh Generation • Robotics Simulation • IoT & Embedded Systems • Hardware-Software Integration

Internship

JULY 2024 – DECEMBER 2025
AI & Robotics Intern – JuzCode Robotics & AI Pvt. Ltd., Kerala (Hybrid)
Contributing to the design, development, and testing of cutting-edge AI and robotics solutions while participating in a comprehensive corporate skill development program.

- ✓ Developed and optimized AI/ML components for robotics projects, including analytical modeling, simulation, and real-time control logic implementation for autonomous systems.
- ✓ Participated in end-to-end project workflows: requirements analysis, prototyping, testing, and technical documentation to ensure solution quality and reliability.
- ✓ Implemented Retrieval-Augmented Generation (RAG) pipelines using LangChain and ChromaDB for context-aware information retrieval.
- ✓ Collaborated with cross-functional teams and mentors to translate business requirements into robust technical solutions with timely delivery and knowledge transfer.
- ✓ Gained hands-on experience in robotics simulation environments and AI integration frameworks, enhancing both technical depth and practical problem-solving skills.

Paid Internship: 18 hours/week

JULY 2022 – AUGUST 2022
Machine Learning Intern – NSIC Technical Service Centre, Delhi (Remote)
Built and deployed Python-based machine learning solutions for technical service optimization and system performance improvement.

- ✓ Developed ML models for predictive analytics and system optimization, applying data-driven approaches to solve real-world technical challenges.

Education

JULY 2024 – JULY 2026 | Master of Engineering – Computer Science and Engineering
Mar Ephraem College of Engineering and Technology, Kanyakumari, Tamilnadu.
- ✓ Coursework: AI, Computer Vision, Robotics, Distributed Systems, Real-time Systems
- ✓ Focus on deep learning applications, real-time processing, and robotics system design
- ✓ Strong academic performance with emphasis on practical project-based learning

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AUGUST 2019 – MAY 2023 | Bachelor of Engineering – CSE (AI & ML Specialization)
Annamalai University, Chidambaram, Tamil Nadu - CGPA: 8.64/10
- ✓ Coursework: Machine Learning, Computer Vision, IoT, Statistics, Data Structures, Programming
- ✓ **Leadership Roles:** Founder & Core Member – Tech Club; Treasurer – CSEA (Computer Science Engineering Association); Member – CSI (Computer Society of India)

Projects

1. AI-Driven Gesture Recognition and Virtual Control System for HCI – JULY, 2026
Technologies: Python • OpenCV • MediaPipe • TensorFlow • TensorFlow Lite • PyAutoGUI • SpeechRecognition • PyTTSx3 • DeepFace • NumPy
- ✓ Developed real-time hand gesture recognition using OpenCV and MediaPipe.
- ✓ Implemented a virtual mouse with cursor control and built gesture-controlled presentation, virtual keyboard, and desktop automation features.
- ✓ Integrated facial recognition, eye tracking, blink detection, and voice commands for intelligent interaction.
- ✓ Optimized the system using TensorFlow Lite for efficient, low-latency real-time performance.

Key Impact: Delivered a scalable AI-based touchless desktop control system that enhances accessibility and improves Human–Computer Interaction.

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2. Visual-Based Virtual Mouse – Hand Gesture Recognition System – NOV, 2025
Technologies: Python • OpenCV • TensorFlow Lite • MediaPipe • NumPy • Scikit-Learn
Built a real-time, production-grade hand gesture recognition system enabling intuitive mouse cursor control and input operations through hand gesture movements without physical contact.
- ✓ Implemented robust real-time hand detection and tracking using computer vision algorithms optimized for low-latency performance in varying lighting conditions and hand orientations.
- ✓ Developed deep learning-based gesture classification supporting multiple gesture types: open hand, pinch grip, pointer, grab—enabling natural cursor movement, clicking, and scrolling operations.
- ✓ Integrated calibration and sensitivity adjustment features, improving usability and user experience for diverse hand sizes and movement patterns.

Key Impact: Demonstrated hands-free computer control with intuitive, natural interaction—addressing accessibility needs and advancing Human-Computer Interaction research.

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3. 2D Images to 3D Triangular Mesh Conversion System – MAY, 2023
Technologies: Python • TensorFlow • Graph Convolutional Neural Networks (GCNN) • OpenCV • NumPy
Developed an intelligent deep learning system to automatically generate high-quality 3D triangular mesh models from 2D images using advanced neural network architectures.
- ✓ Designed and trained Graph Convolutional Neural Networks to learn 3D geometric structures from 2D image inputs across diverse image types and resolutions.
- ✓ Implemented complete ML pipeline: data preprocessing, model training, mesh generation algorithm, and result validation with algorithmic optimisations for computational efficiency.

Certifications

Applied Accelerated Artificial Intelligence – NPTEL / SWAYAM, 2022

Additional Information

Languages: Tamil • English
Hobbies: Music • Singing • Drawing • Sports • Reading Books.

Last Updated: July 2026